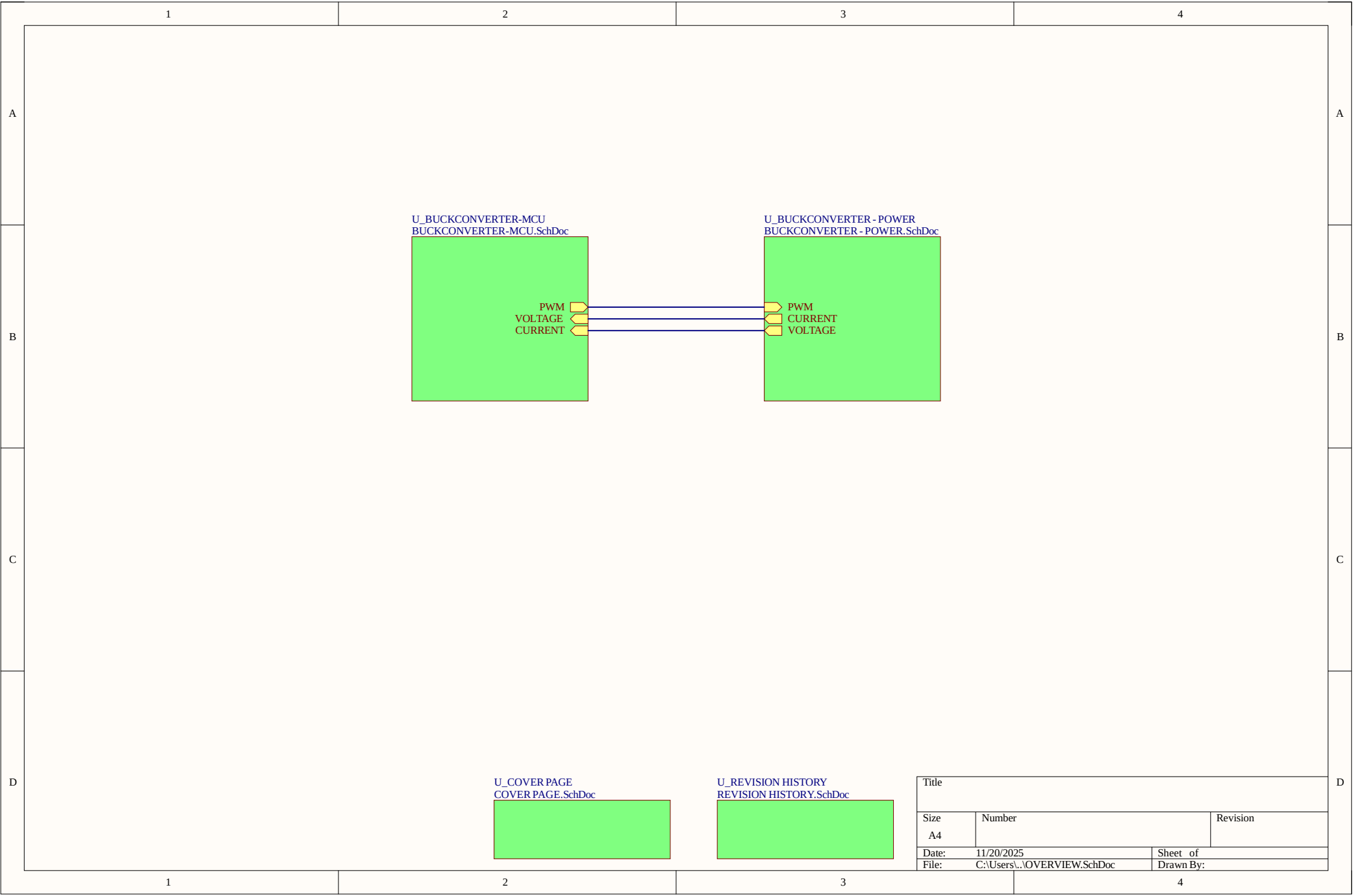




BUCK CONVERTER

11/20/2025

Variant: No variant

DRAFT 20-09-2025

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DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational
design notes.

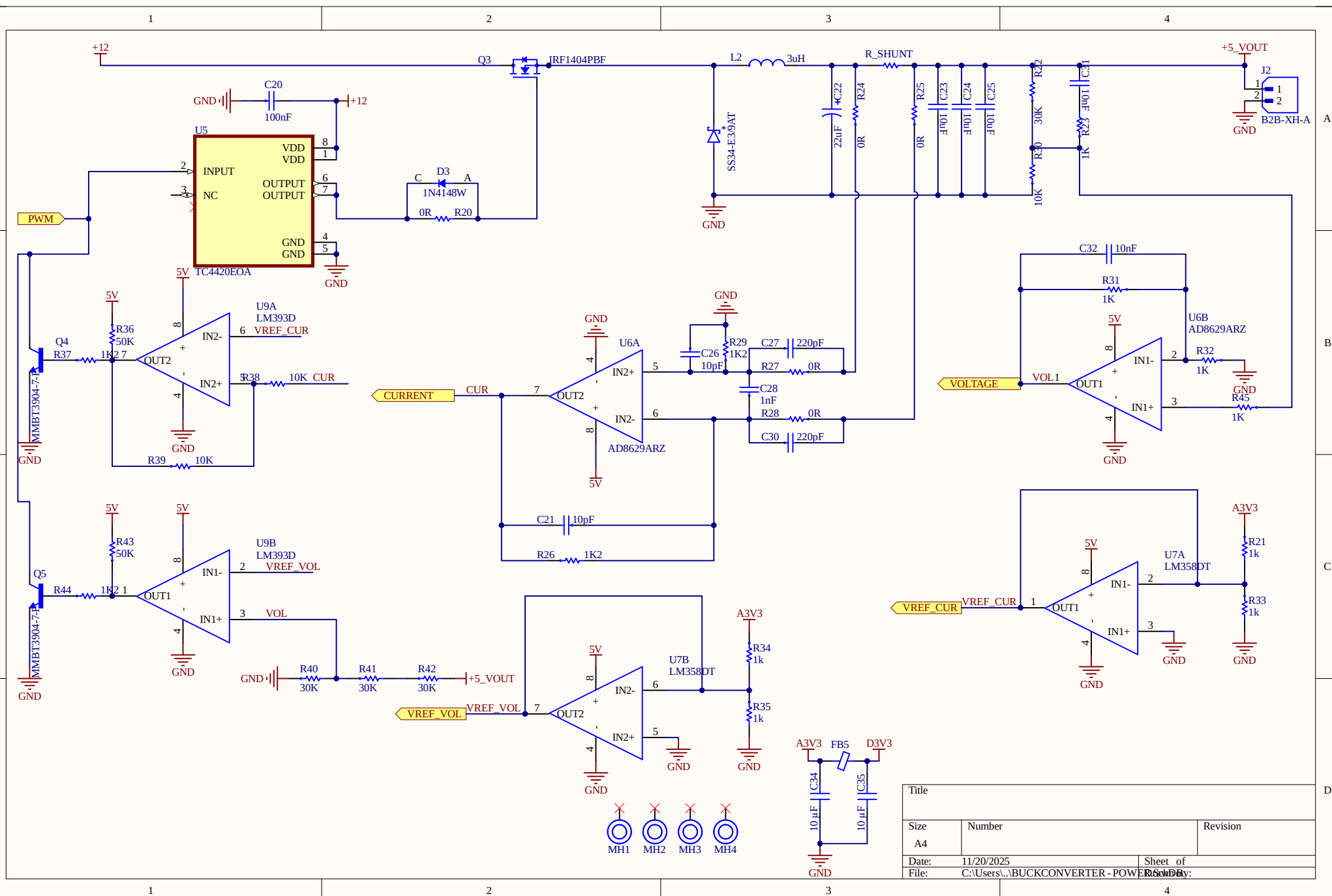
DESIGN NOTE:
Example text for cautionary
design notes.

DESIGN NOTE:
Example text for debug notes.

DESIGN NOTE:
Example text for critical
design notes.

LAYOUT NOTE:
Example text for critical
layout guidelines.

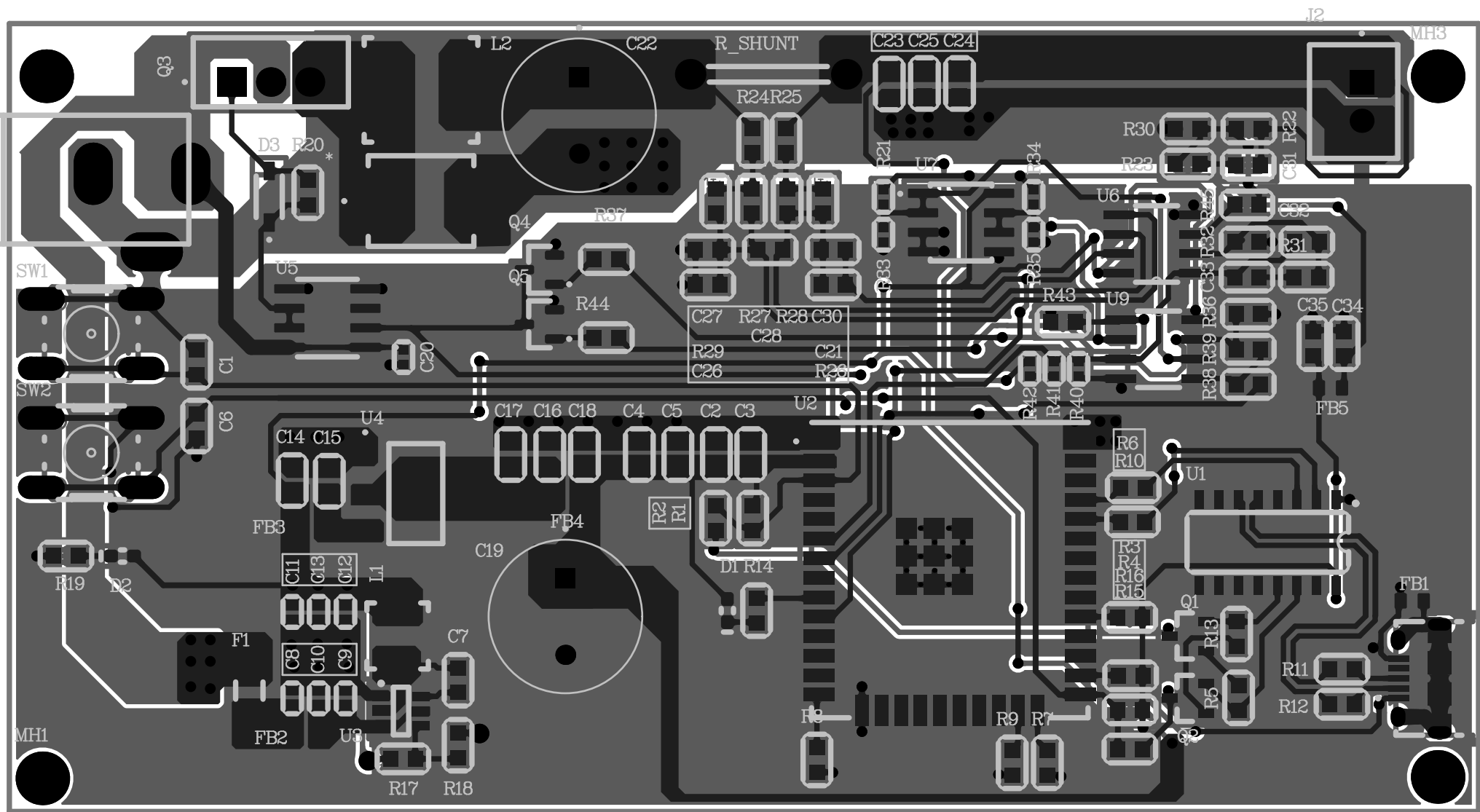
Title		
Size A3	Number	Revision
Date: 11/20/2025	Sheet of	
File: C:\Users\...\COVER PAGE.SchDoc	Drawn By:	



1		2		3		4	
A							A
B							B
C							C
D							D
1		2		3		4	

Title		
Size A4	Number	Revision
Date:	11/20/2025	Sheet of
File:	C:\Users\...REVISION HISTORY.SchDoc Drawn By:	

Title		
Size A4	Number	Revision
Date:	11/20/2025	Sheet of
File:	C:\Users\...\REVISION HISTORY.SchDoc Drawn By:	



Component	Description	Quantity	Footprint	Label	Quantity
1034-ES16AT	Rectifier Diode, Schottky, 1 Phase, 1 Element, 3A, 40V - VRRM, Silicon, DO-29AA		ES16	1034-ES16AT	1
100nF	CAP0603 X7R 100F 10% 25V		C1, C2, C3, C6, C16	CAP 0603	DMC10V7500K10N1
10µF	CAP0603 X7R 100F 10% 25V		C3, C4, C14, C15, C17, C18, C24, C25	CAP 0603	DMC10V7500K10N1
1000µF	CAP CER 1000µF 10% 50V X7R 0603 3MAD		C7	CAP-CER0603-0.8H	CAP0603000 1000F 50V X7R 0603 3MAD 1
100µF	CAP CER 100µF 20% 6.3V X5F 0402 3MAD		C8, C9, C10, C11, C12, C13	CAP-CER0402-20WH	CAP0604002 100µF 50V 6.3V X5F 0402 3MAD
1CA1CH33EL	Cap Aluminum 3300µF 50V 20% (EL) (1.5mm) Radial Aluminum Electrolytic Can 1.5mm 200mA 1000 hr 150C Bulk		C19	3300µF-50mm	1CA1CH33EL
100nF	CAP CER 100nF 20% 6.3V X5F 0402 3MAD		C20	CAP-CER0402-20WH	CAP0604002 100nF 50V 6.3V X5F 0402 3MAD
10pF	CAP CER 220pF 10% 50V X7R 0603 3MAD		C21, C26	CAP-CER0603-0.8H	CAP0606003 220pF 50V X7R 0603 3MAD
10µF			C22	3300µF-50mm	CAP-45µF 50V 1
10µF	SMD Multilayer Ceramic Capacitor 10 µF, 25 V, 0603 [5008 Metric], 5% 20%, X5F		C23, C24, C25	Cap var 0603	07T180F01020M013 0
100pF	CAP CER 220pF 10% 50V X7R 0603 3MAD		C27, C30	CAP-CER0603-0.8H	CAP0606003 220pF 50V X7R 0603 3MAD
10pF	CAP CER 220pF 10% 50V X7R 0603 3MAD		C28	CAP-CER0603-0.8H	CAP0606003 220pF 50V X7R 0603 3MAD
10pF	SMD Multilayer Ceramic Capacitor 10 µF, 25 V, 0603 [5008 Metric], 5% 20%, X5F		C31, C32	Cap var 0603	07T180F01020M013 0
10mm	LED SMD T1835 600nm 6003		C3, C2	LED	LED ORANGE 600NM 0603
2N4548W	Small Signal Diode, Single, 100V, 300mA, Consistent Yes		C3	2N4548	2N4548W 1-F
C15	Fuse Chip Fast Acting 5A 32V SMD 5mmx7mm 1.206 3.2 X 1.58 X 1.58mm		F1	FUSE C15	C15
100 Ohms @ 125 MHz	Thin Film Resistor 100 Ohms 1% 0.125W		R11, R12, R14, R15	FR_0603	RLM30K0325H02
100MHz 2A 50mOhm	Power Resistor Multilayer 100MHz 20% 100mA 2A 50mOhm 50V 0603 2.5mmx7mm		R3	FR 0603	RLM30P1211N10
MC1202-B	MC1202-B		C1	MC1202-USB-B	MC1202-USB-B
MC1202-B	MC1202-B		C2	MC1202-USB-A	MC1202-USB-A
MC1202-B	MC1202-B		C3	MC1202-USB-C	MC1202-USB-C
2.5µH	IND MC1202 2.5µH 20% 5A 0603 3MAD		L1	ND-0402-2R2	2.5µH 20% 5A 0402 3MAD
3µH	IND MC1202 3µH 20% 5A 0603 3MAD		L2	ND-0603-3µH	3µH 20% 5A 0603 3MAD
MOUNTING-HOLE	MH1, MH2, MH3, MH4-M3		MH		
NMOS	MMBT3004 Series 40 V CE Transistor 0.2 A NPN General Purpose Amplifier - BJT - Si		C1, C2	10T23-3	MMBT3004
WFS4049RF	Single N-Channel 40 V 0.8mA Ohm 10W-C HEVLET+ Power 3		C3	TC020 AB	WFS4049RF
MMBT3004 7-F	Transistor General Purpose BJT NPN 40 Volt 0.2A Automotive 3-Pin SOT-23		C4, C5	MMBT3004	MMBT3004 7-F
10K	Res Thick Film 0603 2.2K Ohms 5% 1/10W 4000ppm/C Molded SMD (SMD) Paper T/R		R1, R2, R4, R5, R7, R8, R9, R11, R16	Res 0603	Resistor 2.2K
1K	Res Thick Film 0603 2.2K Ohms 5% 1/10W 4000ppm/C Molded SMD (SMD) Paper T/R		R2, R15, R20	Res 0603	Resistor 2.2K, Resistor 0K
10K	Res Thick Film 0603 2.2K Ohms 5% 1/10W 4000ppm/C Molded SMD (SMD) Paper T/R		R6, R10, R11, R12	Res 0603	Resistor 2.2K
1K	Res Thick Film 0603 2.2K Ohms 5% 1/10W 4000ppm/C Molded SMD (SMD) Paper T/R		R14, R19	Res 0603	Resistor 2.2K
49.9K	RES CAP 207 1% 1/10W 0603 3MAD		R17	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R18, R20, R28, R30	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
1K	RES CAP 207 1% 1/10W 0603 3MAD		R21, R33, R34, R35	RES-CHP0402-0.39H	RESCHP0402 33K 2% 0402 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R22	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
1K	RES CAP 207 1% 1/10W 0603 3MAD		R23, R31, R32, R40	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
1K	RES CAP 207 1% 1/10W 0603 3MAD		R24, R25, R27, R28	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R26, R29, R37, R44	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R36, R43	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R40, R41, R42	RES-CHP0402-0.39H	RESCHP0402 33K 2% 0402 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R43, R44, R45	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R46, R47, R48	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R49, R50, R51	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R52, R53, R54, R55	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R56, R57, R58	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R59, R60, R61	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R62, R63, R64	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R65, R66, R67	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R68, R69, R70	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R71, R72, R73	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R74, R75, R76	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R77, R78, R79	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R80, R81, R82	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R83, R84, R85	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R86, R87, R88	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R89, R90, R91	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R92, R93, R94	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R95, R96, R97	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R98, R99, R100	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R101, R102, R103	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R104, R105, R106	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R107, R108, R109	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R110, R111, R112	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R113, R114, R115	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R116, R117, R118	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R119, R120, R121	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R122, R123, R124	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R125, R126, R127	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R128, R129, R130	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R131, R132, R133	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R134, R135, R136	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R137, R138, R139	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R140, R141, R142	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R143, R144, R145	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R146, R147, R148	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R149, R150, R151	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R152, R153, R154	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R155, R156, R157	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R158, R159, R160	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R161, R162, R163	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R164, R165, R166	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R167, R168, R169	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R170, R171, R172	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R173, R174, R175	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R176, R177, R178	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R179, R180, R181	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R182, R183, R184	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R185, R186, R187	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R188, R189, R190	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R191, R192, R193	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R194, R195, R196	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R197, R198, R199	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R200, R201, R202	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R203, R204, R205	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R206, R207, R208	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R209, R210, R211	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R212, R213, R214	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R215, R216, R217	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R218, R219, R220	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R221, R222, R223	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R224, R225, R226	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R227, R228, R229	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R230, R231, R232	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R233, R234, R235	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R236, R237, R238	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R239, R240, R241	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R242, R243, R244	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R245, R246, R247	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R248, R249, R250	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R251, R252, R253	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R254, R255, R256	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R257, R258, R259	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R260, R261, R262	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R263, R264, R265	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD
10K	RES CAP 207 1% 1/10W 0603 3MAD		R266, R267, R268	RES-CHP0603-0.49H	RESCHP0603 207 2% 0603 3MAD